

Photonic Quantum Error Correction from SEFI/GWFM Geometry

Geometric Stability, Error Surfaces, and Syndrome Structure

Jason Duran Dutton

2026

Abstract

Photonic quantum systems suffer from loss, dephasing, mode mixing, and spectral drift. Conventional photonic quantum error correction (QEC) encodes information in bosonic codes or multi-photon entangled states, but these approaches lack a geometric principle governing error formation and correction. SEFI/GWFM provides such a principle by defining stability surfaces, geometric identity layers, and worldline-based field emergence. This manuscript develops a geometric framework for photonic QEC based on SEFI/GWFM, including stability surfaces, photonic error geometry, geometric stabilizers, syndrome geometry, SPDC geometry, and CHSH geometry. The resulting structure provides a physically grounded foundation for photonic QEC.

1 Introduction

Photonic quantum information processing offers long coherence times, room-temperature operation, and compatibility with integrated photonics. However, photonic systems face unique error channels:

- photon loss,
- dephasing,
- mode mixing,
- spectral wandering.

SEFI/GWFM provides a geometric foundation for photonic QEC by defining identity layers, worldline geometry, and stability surfaces. Errors correspond to geometric displacements off these surfaces, and correction corresponds to restoring geometric sovereignty.

This manuscript develops photonic QEC from SEFI/GWFM.

2 Stability Surfaces

FIELD SOVEREIGNTY defines a stability surface: a geometric manifold representing allowed transformations of the photonic mode. Let S denote the stability surface in identity space.

Errors correspond to displacements orthogonal to S :

$$\Delta \mathbf{I}_\perp(t) \neq 0.$$

Loss, dephasing, mode mixing, and spectral drift correspond to distinct geometric displacements.

Stability surfaces provide a geometric interpretation of photonic error channels.

3 Photonic Error Geometry

Photonic errors correspond to geometric deviations from sovereignty-preserving worldline behavior.

3.1 Loss

Loss corresponds to worldline termination or collapse into boundary regions of the stability surface.

3.2 Dephasing

Dephasing corresponds to torsion-induced geometric phase deviation.

3.3 Mode Mixing

Mode mixing corresponds to displacement into neighboring stability surfaces.

3.4 Spectral Drift

Spectral wandering corresponds to translation along the spectral axis of the stability manifold.

4 Geometric Stabilizers

Geometric stabilizers preserve sovereignty by enforcing curvature, torsion, spectral, and polarization constraints.

A stabilizer G acts on the identity state as

$$G(\mathbf{I}(t)) = \mathbf{I}(t).$$

Stabilizers correspond to geometric invariants:

- curvature constraints,
- torsion constraints,
- spectral-position constraints,
- polarization-surface constraints.

These stabilizers define geometric QEC codes.

5 Syndrome Geometry

Weak or QND photonic measurements sample geometric deviation from stability surfaces.

Syndrome geometry includes:

- curvature deviation,
- torsion deviation,
- spectral displacement,
- polarization rotation.

A syndrome vector is defined as

$$\mathbf{S}(t) = (\Delta K(t), \Delta T(t), \Delta\omega(t), \Delta P(t)).$$

Syndrome geometry provides a geometric method for error detection.

6 SPDC Geometry

Spontaneous parametric down-conversion (SPDC) produces annular momentum-space geometry corresponding to a stability surface defined by FIELD ORIGIN and FIELD SOVEREIGNTY.

Diametric correlations arise from sovereign geometric symmetry:

$$\mathbf{k}_s = -\mathbf{k}_i.$$

SPDC geometry provides a natural geometric stabilizer for photonic QEC.

7 CHSH Geometry

Polarization evolution is treated as a worldline on the Poincaré-Bloch sphere. Bell states correspond to minimal-torsion worldlines.

CHSH correlations arise from geometric invariants of polarization worldlines.

The CHSH inequality corresponds to a sovereignty-preserving constraint on polarization geometry.

8 Discussion and Outlook

SEFI/GWFM provides a geometric foundation for photonic QEC. Stability surfaces define allowed transformations, errors correspond to geometric displacements, stabilizers enforce sovereignty invariants, and syndrome geometry samples deviation.

Future work will focus on:

- deriving explicit photonic stabilizer codes,
- exploring geometric QEC lattices,
- developing integrated photonic implementations,
- coupling SEFI Space to photonic hardware.